Pre-Class Assignment: Chapter 20 (Fairness Handouts)

Independence, Separation, Sufficiency, the Impossibility Result, and a Way Out

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

Reading: Two course handouts: (1) *Algorithmic Fairness Definitions and their Incompatibility*; (2) *A Way Out of the Impossibility: The First-Best Screening Rule*. Optional background: Gans Ch. 20 §20.2–§20.3.

Instructions: Answer each question in 1–3 sentences. These questions are designed to be straightforward if you did the reading.

You must submit this assignment in person. The assignment must be handwritten. No photocopies will be accepted.

Part 1 — Classification metrics

1. In the handout's notation, $e_1 = \Pr[s = 1 \mid \theta = 1]$ and $e_2 = \Pr[s = 0 \mid \theta = 0]$. In plain language, what does e_1 measure, and what does e_2 measure? Give the everyday name (sensitivity / specificity, recall, etc.) for at least one of them.

2. The handout writes precision as

$$\text{Precision} = \frac{e_1\, p}{e_1\, p + (1 - e_2)(1 - p)}.$$

   Why does the base rate p appear in this formula, even though p is a fact about the population and not about the algorithm? Answer in one or two sentences.

Part 2 — The three fairness criteria

3. **Independence** (demographic parity) requires $\Pr[s = 1 \mid G = 1] = \Pr[s = 1 \mid G = 0]$. In plain language, what does this require the algorithm to equalize across groups?

4. **Separation** requires that, *conditional on the true outcome* θ, the prediction is independent of group. In the handout's (e_1, e_2) notation, this means $e_1(1) = e_1(0)$ and $e_2(1) = e_2(0)$. Why does the handout say this is the criterion most closely aligned with what non-technical audiences mean by "the algorithm doesn't discriminate"?

5. **Sufficiency** (predictive parity) requires that, *conditional on the prediction* s, the true outcome is independent of group. In one or two sentences, why is this the criterion most familiar to actuaries, doctors, and bankers?

Part 3 — The impossibility result

6. The handout's "standing assumptions" require $e_1 + e_2 > 1$ and $e_2 < 1$. In one sentence each, what do these two assumptions rule out?

7. State the theorem in your own words: under the standing assumptions, what does "any two of {independence, separation, sufficiency}" imply about the base rates $p(0)$ and $p(1)$?

Part 4 — Picking among the three

8. Suppose a policymaker insists on **independence** (demographic parity). According to the handout's table, what *two* other things must she give up when base rates genuinely differ across groups?

Part 5 — A way out of the impossibility (first-best handout)

9. The social welfare function in the handout is

$$\text{SWF} \;=\; \sum_{i=1}^{N} \psi_{G_i}\, \theta_i\, a_i.$$

   In one sentence, where in this formula does the planner's equity preference live — on the truth θ_i, on the decision a_i, or somewhere else?

10. The handout claims that "no AI score appears anywhere" in the SWF, and that "the score will emerge as soon as we take the expectation." In one sentence, what does the **law of iterated expectations** produce when applied to $\mathbb{E}[\sum_i \psi_{G_i}\, \theta_i\, a_i]$, and what is the name the handout gives to the object that emerges?

11. The handout defines the score as

$$s_i \;:=\; \Pr[\theta_i = 1 \mid G_i, X_i].$$

    In one sentence, why does the conditioning set in this definition include **both** G_i and X_i, and what would happen to s_i if you dropped G_i?

12. State the **first-best admissions rule** in one line, using the handout's notation. (You only need to write the formula for a_i^*.)

13. The handout argues that the equity weight ψ_g does *not* enter the threshold comparison $s_i > t(G_i; K)$ directly. Instead, ψ_g controls something else. In one sentence, what does ψ_g actually determine in the solution?

14. State the **equity-irrelevance result** in one sentence, in your own words.